

Answer key

Version A

Marking guide and worked solutions · 40 points · instructor copy: do not distribute before the exam

How to mark. Part 1 and the Summary reward **interpretation that is correct, specific and clearly explained in the student's own words**; there is no single right model. A full coefficient interpretation has: the change in X **with its unit** (1 °C, €0.10, 10 km/h); the change in predicted Y **with direction and unit** (more or fewer cups); **"holding the other variables constant"** (naming them); and association language ("is associated with", "the model predicts"), not "causes". For 0/1 variables the comparison group must be clear (event vs no event). Apply **follow-through** from earlier answers; accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Students interpret coefficients but are **not yet** asked whether coefficients are significant. In Part 3 a t-test is significant if $p < 0.05$ (two-tailed, T.TEST type 2). Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: temperature (up: hot days want cold treats), weekend (up: more leisure time), beach event (up: crowds nearby), planned price (down: dearer gelato sells less), sunshine (up, but overlaps with temperature).	3	1 per input: direction and a coherent reason.
1b	LO5 · Con	Item 6 , iced coffee sales counted at closing time: it is not known before opening and it is driven by the same weather and day type, so it cannot be used to forecast.	2	1 item 6 + 1 the problem (not known in advance).
1c	LO3, LO5, LO6 · Con	Not both. With $r = 0.78$ they carry mostly the same information, so the model cannot tell their effects apart, and "holding sunshine constant while temperature changes" is unrealistic because they rise and fall together: the coefficients become unreliable. Keep temperature ($r = 0.58$ with sales vs 0.49; a more direct measure of heat).	2	1 overlap and why it is a problem + 1 a choice with a reason.
2a	LO5, LO6 · Com	Sales = $167.59 + 7.06 \times \text{temperature} + 30.65 \times \text{event} - 33.73 \times \text{price}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; Table 2: Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 numbers matched to the right variables + 1 equation form with variable names.
2b	LO6, LO7 · Con	(i) Holding beach event and price constant, a 1 °C rise in temperature is associated with about 7 more gelato cups sold per day (7.06). (ii) The other variables also change sales and may differ between hot and cooler days. Holding them constant isolates the effect of this variable: we compare days that are alike in the other variables and differ only in this one.	3	(i) 1 size, direction and unit + 1 names the variables held constant; (ii) 1 reason.
2c	LO6, LO7 · Con	(i) Negative: higher price goes with lower sales : each €1 increase is associated with about 34 fewer cups, holding temperature and event constant. (ii) On event days the model predicts about 31 more cups than on days without an event, at the same temperature and price. (iii) $0.10 \times (-33.73) = -3.37$: about 3 fewer cups per €0.10 increase .	3	1 sign meaning in context + 1 0/1 variable with the comparison group + 1 rescaled coefficient.
2d	LO3, LO6, LO7 · Cal + Con	(i) $10 \times 7.06 = 70.6$: about 71 more cups per 10 °C . (ii) "Per 10 °C" is better: 1 °C is too small to picture, while hot and cool days differ by 10 °C or more; choose a change that matches realistic variation. (iii) Not fair: the coefficients are per unit of different variables (event is a 0/1 switch, temperature is per °C). A 10 °C rise (+71) is bigger than an event (+31), so temperature matters more for realistic changes.	3	1 rescaled + 1 why a sensible unit + 1 unfair comparison explained with (i).
2e	LO5, LO6, LO7 · Cal + Con	(i) $\hat{y} = 167.59 + 7.06(26) + 30.65(1) - 33.73(4.30) = \mathbf{236.8 \text{ cups}}$. (ii) $0.20 \times (-33.73) = -6.75$: about 7 fewer cups .	3	2 prediction (1 substitution, 1 value) + 1 price change effect with sign.
3	LO3, LO4, LO6 · Con	(a) Day 8 : the hottest day, but sales far below the pattern. (b) Without it the slope is 6.29 vs 2.49 with it, R^2 0.97 vs 0.17: one day cuts the slope by more than half and wrecks the fit. (c) Find out why (closure, power cut, stock-out, recording error). Delete only with a documented reason; otherwise keep it and report results both ways.	3	1 + 1 + 1.
4a	LO3 · Cal	Means 200 and 186; $\Sigma(x - \bar{x})^2 = 320$ and 256; $s_1 = \sqrt{(320 \div 4)} = 8.94$, $s_2 = \sqrt{(256 \div 4)} = 8.00$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.

4b	LO3 · Cal	$s_p = \sqrt{[(4 \times 80 + 4 \times 64) \div 8]} = \sqrt{72} = 8.49$; $t = (200 - 186) \div [8.49 \times \sqrt{(1/5 + 1/5)}] = 14 \div 5.37 = \mathbf{2.61}$.	2	1 s _p + 1 t (follow-through on s; either sign if the direction is stated).
5a	LO3, LO5 · Com	(i) =STDEV.S(B2:B6) (or STDEV) · (ii) =T.TEST(B2:B6, C2:C6, 2, 2) · (iii) =LINEST(B2:B169, C2:E169, TRUE, TRUE). Sheet names are not required.	3	1 each (correct function and ranges; arguments in the right order).
5b	LO3, LO5 · Com	In T.TEST the first 2 means a two-tailed test and the second 2 means two samples with equal variance . In LINEST the first TRUE keeps the intercept (not forced to 0); the second TRUE returns the extra statistics (standard errors, R ² , SE – Y, F, df, sums of squares) in rows 2–5.	2	1 T.TEST arguments + 1 second TRUE.
5c	LO1, LO3, LO6 · Con	(i) H ₀ : the price rise made no difference to mean daily gelato sales (mean before = mean after). (ii) p = 0.031 < 0.05, so the difference is statistically significant : reject H ₀ . (iii) If the price had no effect, a difference this large or larger would turn up in only about 3% of samples like these. It is not the probability that H ₀ is true and it does not give the size of the effect.	3	1 null hypothesis + 1 decision using p + 1 correct plain-words meaning.
6	LO7 · Con	(i) One model input with a sensible change and its effect on sales, holding the others constant (e.g. “Holding event and price the same, a 10 °C warmer day is expected to sell about 71 more gelato cups.”). (ii) The price test in plain words: gelato sales were lower after the price rise (about 14 cups, p = 0.031). (iii) A caution (small samples, other changes, association not proof, forecasts can miss).	3	1 + 1 + 1; plain language and correct units.

2 · Worked table (Part 2)

Day	Before (x ₁)	x ₁ – $\bar{x}_1$	(x ₁ – $\bar{x}_1$) ²	After (x ₂)	x ₂ – $\bar{x}_2$	(x ₂ – $\bar{x}_2$) ²
1	196	–4	16	184	–2	4
2	204	+4	16	192	+6	36
3	188	–12	144	176	–10	100
4	212	+12	144	196	+10	100
5	200	0	0	182	–4	16
Sum	1000	0	320	930	0	256
Mean $\bar{x}$	200			186		
s	8.94			8.00		

Check against Sheets: =T.TEST gives p = 0.0312 (df = 8, |t| = 2.609, two-tailed).

3 · Where students go wrong

- **Leaving out “holding the other variables constant”** (2b, 2c, Q6), or not naming what is held constant.
- **Treating a coefficient as the total effect of a big change** (2c(iii), 2d): the coefficient is per **one unit**; students must multiply by the change they mean (€0.10, 10 °C, 10 km/h).
- **Sign confusion**: a negative coefficient means sales **fall** as X rises; it does not mean negative sales or “no sales”.
- **Comparing raw coefficients across variables with different units** (2d(iii)).
- **Reading the LINEST table backwards** (2a): Col A is the **last** X column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Causes” and “per customer”**: the coefficients describe daily totals and associations.
- **Reading p as the probability that H₀ is true** (5c). Credit “if there were no real effect, a result this extreme would be rare”.